

The Theorem Ladder of Resolution Geometry

A Consolidated Structural Leg: What Is Forced, Declared, Imported, and Open,

from the Kernel-Pair Root to the Observable Cone

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Abstract

The corpus has grown into many separate hostile-attack papers—kernel pair, quotient, assembly, reconstruction, ignorance, observation limit, saturation, cone, scale, transport, duration. This note supplies the missing independent leg: a single dependency-ordered ladder that states, for the whole framework, exactly what forces what. It introduces no new theorem. Its content is the *typing*: every rung is marked **forced** (by the exact kernel pair), **imported** (established mathematics reused), **gated** (valid only under an explicit hypothesis), **declared** (added structure, not forced), **no-go** (a forced negative), or **open** (a conjecture/gap). The spine is: an observation map F forces its kernel pair R_F , and R_F forces—and forces *only*—the exact extensional layer: observational equivalence, the quotient $Q_F \cong \text{Im } F$, fibre saturation, and universal factorization of fibre-constant constructions. Every higher structure crosses a type boundary and needs its own gate: the tangent shadow dF (regularity), the second-order covariance/Fisher shadow (a declared metric; Fisher provably insufficient), the statistical order (Blackwell $\neq$ kernel-pair order), assembly completeness (a sound groupoid need not generate every fibre), the observable cone (a declared fibre-compatible reachability plus a topology and a Lorentzian-realization gate, after which HKMM reconstruction gives a conformal class), scale (an integrable local clock field; a single worldline underdetermines it), and transport (the induced connection on regular strata, with the canonical connection the largest open gap). The kernel-pair gap is therefore mapped precisely: the root is universal for exact information and for nothing else, and each extension is a named module with a named hypothesis. The result is a leg one can audit rung by rung, with the frontier—connection reconstruction, Blackwell realizability, singular strata, and the physical scale anchor—marked as such.

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1 Why a leg, and how to read it

Each existing paper attacks one joint. What a foundation additionally needs is a spine: the ordered list of load-bearing steps, each tagged by *what kind* of statement it is, so a reader can see where mathematics forces the structure, where a hypothesis is silently doing the work, and where the frontier lies. The tags are:

forced	compelled by the exact kernel pair R_F alone
imported	established mathematics reused, not RG-original
gated	valid only under an explicitly stated hypothesis
declared	added structure, chosen not forced
no-go	a forced negative result (something cannot be done)
open	conjecture or gap; not established

The organizing fact, established by the kernel-pair-root attack, is that only the first stratum below is *forced*; every later rung crosses a type boundary.

2 The ladder

Rung	Statement	Type	Depends on
<i>Stratum 0 — the exact root (the only forced layer)</i>			
R0	Observation map $F : X \rightarrow Y$ declared	input	—
R1	Kernel pair $R_F = X \times_Y X$ is an equivalence relation (exact ignorance)	forced	R0
R2	Quotient $Q_F = X/R_F \cong \text{Im } F$; every fibre-constant map factors uniquely through Q_F	forced	R1
R3	Saturation: observable invariants are exactly R_F -invariant functions on Q_F	forced	R2
<i>Stratum 1 — assembly (bottom-up = top-down, conditionally)</i>			
R4	Kernel groupoid $K_{\text{obs}} = X \times_{Q_F} X$; R_F is a groupoid	forced	R1
R5	Declared automorphism groupoid A ; sound $A \hookrightarrow K_{\text{obs}}$ gives surjection $X/A \rightarrow Q_F$	gated (soundness)	R4
R6	$X/A \cong Q_F$ iff A is gauge-complete (generates every fibre)	gated (completeness)	R5
<i>Stratum 2 — second-order shadow (metric detection)</i>			
R7	Regular reconstruction: (R, N, Σ) is a canonical gauge-equivariant function of (J, Σ, g)	imported	R2
R8	Fisher $= J^\top \Sigma^{-1} J = \text{inverse Schur} = \text{second-order tangent shadow}$	imported	R7
R9	Fisher insufficiency: blind to the null block at $C_{RN} = 0$; does not reconstruct the covariance	no-go	R8
R10	Tangent caveat: $\ker dF = \text{tangent of the fibre only on regular strata; higher contact is jets of } F, \text{ not covariance}$	gated (regularity)	R2, R8

Rung	Statement	Type	Depends on
<i>Stratum 3 — ignorance and the observation limit</i>			
R11	On regular models $\text{rad } g^F = \ker dF$ (Fisher detects, does not define, ignorance)	gated (regularity)	R8,R10
R12	Persistent null $\mathcal{I}_x = \ker dF_\infty$ over a compatible sensor family; attained finitely (finite dim)	gated (compatibility)	R11
R13	Three layers: sensor = wall $\mathcal{I}_x$; precision (evaluation) and representation (downstream) = removable	forced (typing)	R12
R14	Kernel-pair order $\neq$ Blackwell order; equal identifiability kernel pairs need not be Blackwell-equivalent	no-go	R1
<i>Stratum 4 — declared structure on the quotient</i>			
R15	Fibre-compatible reachability $\preceq$ descends to a preorder on Q_F ; antisymmetrize to a causal order	declared	R2
R16	Future sets J_{obs}^+ ; a cone boundary requires a supplied compatible topology	gated (topology)	R15
R17	Lorentzian-realization gate ($\dim > 2$, distinguishing) $\Rightarrow$ HKMM $\Rightarrow$ conformal class $[g_{\text{obs}}]$	imported (gated)	R16
R18	Scale: $\mathcal{T} = -\log C$ additive on the zero-memory stratum; a single worldline underdetermines the conformal factor	gated	R17
R19	Metric representative exists iff local scale data glue as an exact cocycle (zero scale-holonomy)	gated (integrability)	R18
R20	Cross-stratum (astronomy): remote null test after a correspondence map and nuisance model; ratio-only identifiability	gated	R17,R19
<i>Stratum 5 — transport, and the main open gap</i>			
R21	On a regular metric stratum with projector P : $\nabla^W = P \nabla^{\text{LC}}$; second form $\text{II} = (I - P) \nabla^{\text{LC}}$	imported (gated)	R7,R17
R22	Entropy no-go: $H(X), H(Q), H(X \mid Q)$ are fibre-automorphism invariant; entropy cannot select a connection	no-go	R2
R23	Canonical connection from (J, Σ, K) (memory kernel required): $(J, \Sigma, K) \rightarrow [\nabla]$	open	R21,R22
<i>Stratum 6 — duration algebra (self-contained)</i>			
R24	$\mathcal{T} = -\log C$; memory = BCH defect $\frac{1}{2}[\mathcal{T}, \mathcal{T}] + \dots$; additive iff commuting	imported	R18
R25	Scalar clocks = trace character $\tau = a \text{tr } \mathcal{T}$, unique up to unit (Cauchy)	imported	R24

3 The kernel-pair gap, stated exactly

The single most important structural fact is where the root stops.

Gap 1 (What R_F forces, and what it does not). The exact kernel pair forces exactly R1–R4: observational equivalence, the quotient, fibre saturation, universal factorization, and the kernel groupoid. It forces *nothing else*. Each of the following is a separate type and requires its own hypothesis:

- (i) the *tangent shadow* $\ker dF$ equals the fibre tangent only on regular strata (R10); at singular points it does not.
- (ii) *higher contact* is governed by jets of F , not by any covariance; formal flatness need not imply exact local constancy (R10).
- (iii) *Fisher/covariance* is a declared distinguishability structure, not kernel-pair data, and is insufficient to reconstruct the covariance (R8–R9).
- (iv) the *statistical order* (Blackwell/deficiency) is strictly finer than sameness of identifiability kernel pair (R14).
- (v) *assembly* closes only under gauge completeness; a sound groupoid need not generate every fibre (R6).
- (vi) *precision* and *representation* act on evaluation and on downstream constructions, not on R_F (R13).

These six crossings are the kernel-pair gap. Naming them is the leg’s central service: it forbids smuggling any higher module into the root.

Gap 2 (The main open rung). R23—a canonical connection from (J, Σ, K) —is the largest remaining source of non-uniqueness in the characterized object. The entropy no-go (R22) shows a natural candidate selector fails; the memory kernel (R24) is the extra datum the reconstruction needs; the theorem is not yet in hand. Everything downstream of a connection (curvature, holonomy, stratified transport) inherits this gap.

Gap 3 (The physical anchor is outside the ladder). Absolute scale (R18–R19) and the identification $v_\partial = c$ are declared/empirical, not forced by any rung. This is why the cone yields $[g]$ but not g : the ladder is conformal-complete and scale-open by construction. The anchor is a physical input, correctly placed outside the forced spine.

4 The minimal “must” path to the observable cone

Extracting only the load-bearing rungs, the shortest sound route from an experiment to a conformal observable cone is:

$$\underbrace{F}_{R0} \rightarrow \underbrace{R_F \rightarrow Q_F}_{R1-R2, \text{ forced}} \rightarrow \underbrace{\preceq_{\text{obs}}}_{R15, \text{ declared}} \rightarrow \underbrace{\text{antisymmetrize}}_{R15} \rightarrow \underbrace{\text{topology}}_{R16, \text{ gate}} \rightarrow \underbrace{\text{Lorentzian realization}}_{R17, \text{ gate}} \rightarrow [g_{\text{obs}}].$$

Groupoid assembly (R4–R6) is an *alternative certificate* for R2, not an extra requirement: it proves the same quotient bottom-up, and adds value only by exhibiting the gauge that generates the fibres. The second-order stratum (R7–R11) is required only if one wants the metric/ignorance geometry rather than the bare cone. Scale (R18–R20) and transport (R21–R23) are required only for a metric representative and for dynamics, respectively. This is the honest minimal dependency: three forced rungs, one declared order, two gates, and an imported reconstruction theorem.

5 Non-claims

Non-claim 1 (No new theorem). This leg consolidates and types existing results; it proves nothing new. Its claim is the dependency structure and the exact location of each gate and gap.

Non-claim 2 (No new primitive). The root is a kernel pair; the imported rungs are Schur/Fisher, HKMM reconstruction, BCH, and Cauchy's equation. No rung introduces a primitive unknown to standard mathematics.

Non-claim 3 (Conformal, not metric; observational, not dynamical). The forced and gated rungs reach a conformal class and an observational transport at most. Absolute scale, the light-speed identification, and Einstein dynamics are outside the ladder.

Non-claim 4 (Gaps are gaps). R14 (Blackwell), R23 (canonical connection), singular-stratum reconstruction, and the physical anchor are open or external and are not asserted as closed.

6 Conclusion

The corpus has an independent leg: a spine of twenty-five rungs in seven strata, each typed, with the exact kernel pair forcing only the first four and every later structure carrying a named hypothesis. The observable cone is reached by a short forced-plus-gated path; the metric, the transport, and the physics are further modules, each gated or open. The two structural facts worth carrying are that the kernel pair is universal for exact information *and nothing else*—the six type crossings are the gap—and that the single largest remaining non-uniqueness is the canonical connection R23. A framework that can state its own dependency graph, its imports, and its open rungs this precisely is one whose claims can be audited; that auditability, not any new theorem, is what this leg supplies.

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